D2.3 Water potential

NAME	DATE	TOTAL
		_____ / 36

About 35 minutes, or one section at a time. You will need a calculator. The number of marks shows how many separate points to make.

Section A: Recall

6 MARKS

1 **DEFINE** solvation. [D2.3.1] [1]

2 **DEFINE** hypertonic. [D2.3.2] [1]

3 **STATE** what happens to an animal cell in a hypertonic solution. [D2.3.5] [1]

4 **STATE** the name of the process in which a plant cell membrane pulls away from its wall. [D2.3.6] [1]

5 **STATE** two medical uses of isotonic solutions. [D2.3.7] [2]

Section B: Understanding

14 MARKS

6 **EXPLAIN** why sodium chloride dissolves in water. [D2.3.1] [3]

7 **EXPLAIN** what is happening to water molecules when a cell is in an isotonic solution. [D2.3.3] [2]

8 **COMPARE AND CONTRAST** the effects of a hypotonic solution on animal cells and plant cells. [3]
[D2.3.5, D2.3.6]

ANIMAL CELL	PLANT CELL

9 **EXPLAIN** why a freshwater unicellular organism needs a contractile vacuole. [D2.3.5] [3]

10 **EXPLAIN** why an organ for transplant is bathed in an isotonic solution. [D2.3.7] [3]

Section C: Working with data

11 MARKS

All data in this section are invented for practice. The patterns are realistic.

11 [D2.3.4]

Potato cylinders were placed in sucrose solutions for one hour. Three repeats were carried out.

SUCROSE / MOL DM^{-3}	STARTING MASS / G	FINAL MASS / G
0.0	4.00	4.60
0.2	3.80	4.10
0.4	4.20	4.11
0.6	4.00	3.68

a **CALCULATE** the percentage change in mass at 0.0 and at 0.6 mol dm^{-3}. [2]

b **EXPLAIN** why percentage change is used. [1]

c **ESTIMATE** the isotonic concentration, giving a reason. [2]

d **EXPLAIN** the change in mass at 0.6 mol dm^{-3}. [2]

e **SUGGEST** two ways to make the results more reliable. [2]

f State what standard deviation would show about the repeats. [2]

Section D: Longer answer

5 MARKS

12 EXPLAIN how the movement of water by osmosis affects plant cells, and how this helps to support a plant. [D2.3.3, D2.3.6] [5]